

Lakshay Varshney

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LinkedIn |; GitHub |; Portfolio

Professional Summary

Computer Engineering undergraduate with strong foundations in Data Structures Algorithms, Object-Oriented Programming, and software development. Proficient in C++, Python, and SQL, with hands-on experience building backend services, REST APIs, and end-to-end applications using FastAPI, Docker, and Git. Experienced in integrating AI, LLMs, and RAG pipelines into practical software solutions, with a strong interest in scalable applications, system integration, automation, and enterprise software development. Eager to contribute to software and digital transformation projects while building reliable, production-oriented solutions.

Education

- Aligarh Muslim University** July 2023 – July 2027
Bachelor of Technology (B.Tech) in Computer Engineering Aligarh, India
– CGPA: **8.57/10**
- Anglo Vedic Convent School** April 2022
12th CBSE Board Bahjoi, India
– Grade: **94%**
- Anglo Vedic Convent School** April 2020
10th CBSE Board Bahjoi, India
– Grade: **92.4%**

SKILLS

Programming: Python, C++, SQL

Computer Science: Data Structures & Algorithms, Object-Oriented Programming

AI & LLM Engineering: LLMs, Generative AI, RAG, NLP, Embeddings, Hugging Face, Open-Weight Models, Model Evaluation, Inference

AI/ML: PyTorch, Scikit-learn, Pandas, NumPy, Machine Learning

Backend & APIs: REST APIs, FastAPI, Pydantic, API Development, Software Integration

Databases & Retrieval: SQL, SQLite, MongoDB, FAISS, Qdrant, Vector Databases, Data Processing

DevOps & Cloud: Git, GitHub, GitHub Actions, CI/CD, Docker, AWS, Deployment

Projects

- AI Video Assistant**
Python, Whisper, LLMs, RAG, LangChain, Streamlit
 - Built an AI-powered video analysis assistant that processes video/audio content and converts speech into text using **Whisper** for downstream AI analysis.
 - Implemented an end-to-end pipeline covering **audio extraction, speech-to-text transcription, text processing, and LLM-based content analysis** for automated video understanding.
 - Integrated **Retrieval-Augmented Generation (RAG)** with **LangChain** to enable context-aware question answering over processed video transcripts.
 - Developed an interactive **Streamlit** application allowing users to upload video content, generate AI-powered summaries and insights, and query transcript knowledge using natural language.
- Mental Health Predictor**
Python, Pandas, NumPy, Scikit-learn, FastAPI, Pydantic, REST API
 - Developed an end-to-end **machine learning inference system** using Python and Scikit-learn, transforming raw inputs into structured model-ready features.
 - Implemented **data preprocessing, validation, and feature engineering** to improve consistency and reliability of inputs before model inference.
 - Built modular **FastAPI** and **Pydantic** components for request validation, data processing, model inference, and structured JSON responses.
 - Evaluated multiple model configurations using **GridSearchCV** and selected Random Forest, achieving **86% test accuracy**.
 - Deployed the complete ML inference service and integrated the prediction workflow into a publicly accessible web application. ““latex
- HR Policy Assistant – RAG**
Python, LangChain, Groq, Jina Embeddings, Qdrant, LangSmith, Streamlit
 - Built a **Retrieval-Augmented Generation (RAG)** chatbot for answering employee queries from HR policy documents using **LangChain** and a Groq-hosted LLM.
 - Implemented an end-to-end document pipeline covering **document loading, chunking, Jina embeddings, vector storage, semantic retrieval, and context-grounded response generation**.
 - Integrated **Qdrant Cloud** as the vector database with top-*k* retrieval to provide relevant HR policy context to the LLM and reduce unsupported responses.
 - Implemented **input/output safety guardrails** to detect prompt injection, unauthorized data requests, PII leakage, and unsafe model responses, with **LangSmith** tracing for observability. ““

Certifications

- **Data Structures and Algorithms in C++** — Physics Wallah (2024)
- **Data Science Professional Certificate** — Udemy (2025)
- **Data Analysis Job Simulation** — Deloitte (2025)
- **Remote Sensing and Data Analytics** — ISRO (2025)